

Chenrui Tie

Email: chenrui.tie@u.nus.edu | Website: <https://crtie.github.io/>

EDUCATION

National University of Singapore, Singapore

PhD Student in Computer Science

Aug 2024 - present

Peking University, Beijing, China

Bachelor of Intelligence Science and Technology

Sept 2019 - June 2024

SELECTED PUBLICATIONS

* denotes equal contribution

- [1] Jinxuan Zhu*, **Chenrui Tie***, Xinyi Cao*, Lin Shao, et al. "AdaptPNP: Integrating Prehensile and Non-Prehensile Skills for Adaptive Robotic Manipulation" IEEE International Conference on Robotics and Automation (ICRA) 2026 [\[Project\]](#) [\[Paper\]](#)
- [2] **Chenrui Tie***, Shengxiang Sun*, Lin Shao, et al. "Manual2Skill++: Connector-Aware General Robotic Assembly from Instruction Manuals via Vision-Language Models" IEEE International Conference on Robotics and Automation (ICRA) 2026 [\[Project\]](#) [\[Paper\]](#)
- [3] **Chenrui Tie***, Shengxiang Sun*, Lin Shao, et al. "Manual2Skill: Learning to Read Manuals and Acquire Robotic Skills for Furniture Assembly Using Vision-Language Models" Robotics: Science and Systems(RSS) 2025 [\[Project\]](#) [\[Paper\]](#)
- [4] **Chenrui Tie***, Yue Chen*, Ruihai Wu*, Hao Dong, et al. "ET-SEED: Efficient Trajectory-level SE(3) Equivariant Diffusion Policy " International Conference on Learning Representations(ICLR) 2025 [\[Project\]](#) [\[Paper\]](#)
- [5] Ruihai Wu*, **Chenrui Tie***, Hao Dong, et al. "Leveraging SE(3) Equivariance for Learning 3D Geometric Shape Assembly" International Conference of Computer Vision (ICCV) 2023 [\[Project\]](#) [\[Paper\]](#)

RESEARCH EXPERIENCE

Research on Robot Learning and Intelligent System Lab

PhD student, Advisor: [Prof. Lin Shao](#)

National University of Singapore

Jul 2024 - present

Research on HyperPlane Lab

Undergraduate Research Assistant, Advisor: [Prof. Hao Dong](#)

Peking University

Dec 2022 - Jun 2024

Research on Embodied Perception and Interaction (EPIC) Lab

Undergraduate Research Assistant, Advisor: [Prof. He Wang](#)

Peking University

Aug 2021 - Nov 2022

TECHNICAL SKILLS

- Programming:** Python, C++, Matlab
- Simulators:** SAPIEN, ISAAC SIM, ISAAC GYM, Nimble, DiffCloth, Open Dynamics Engine
- Physical Robot Experience:** Franka, Flexiv rizon, Xarm, LeapHand, Xhand
- Deep Learning Framework:** PyTorch

HONORS AND AWARDS

- | | |
|--|------|
| Research Achievement Award, National University of Singapore | 2026 |
| Research Scholarship, National University of Singapore | 2024 |
| Lee Wai Wing Scholarship, Peking University | 2023 |
| Merit Student award, Peking University | 2023 |
| Merit Student award, Peking University | 2022 |
| Achievement Improvement award, Peking University | 2021 |
| Qin-Jin Scholarship, Peking University | 2020 |